THE UNIVERSITY OF ALABAMA IN HUNTSVILLE | COLLEGE OF SCIENCE

Research and Creative Experience for Undergraduates (RCEU) Program 2025

Doppler Radar Observations of Near-Surface Wind Profiles During the Landfall of Hurricane Laura (2020)

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Motivations

Accurately estimating near-surface winds during hurricane landfalls is crucial for risk assessment and post-storm recovery operations. However, the formation of an internal boundary layer (IBL) overland due to a sudden increase of surface roughness during landfall challenges this effort.

Data and Methods

- Analysis Period: 0027-0553Z on Aug 27, 2020
- KLCH WSR-88D Radar and ASOS (1-min) Data from Lake Charles, LA (KLCH)
- 10 Hz FCMP Tower Data (Balderrama et al. 2011)

Use LROSE RadxConvert and Solo3 to format and quality control radar data

Use Py-ART Velocity Azimuth Display (VAD) package to derive vertical profile of horizontal winds (90-1845 m)

Derive 10-m winds using 90 m and 500 m VAD or beam height winds and the log law (left eq.)

$$V_{10proj} = V_{obs}\frac{\ln(\frac{10}{z_0})}{\ln(\frac{z_{obs}}{z_0})}$$

Log-Law (Leffler et al. 2025)

Z_0 = Surface Roughness Length
Z_{obs} = Observed Wind Height
V_{obs} = Observed Wind Speed

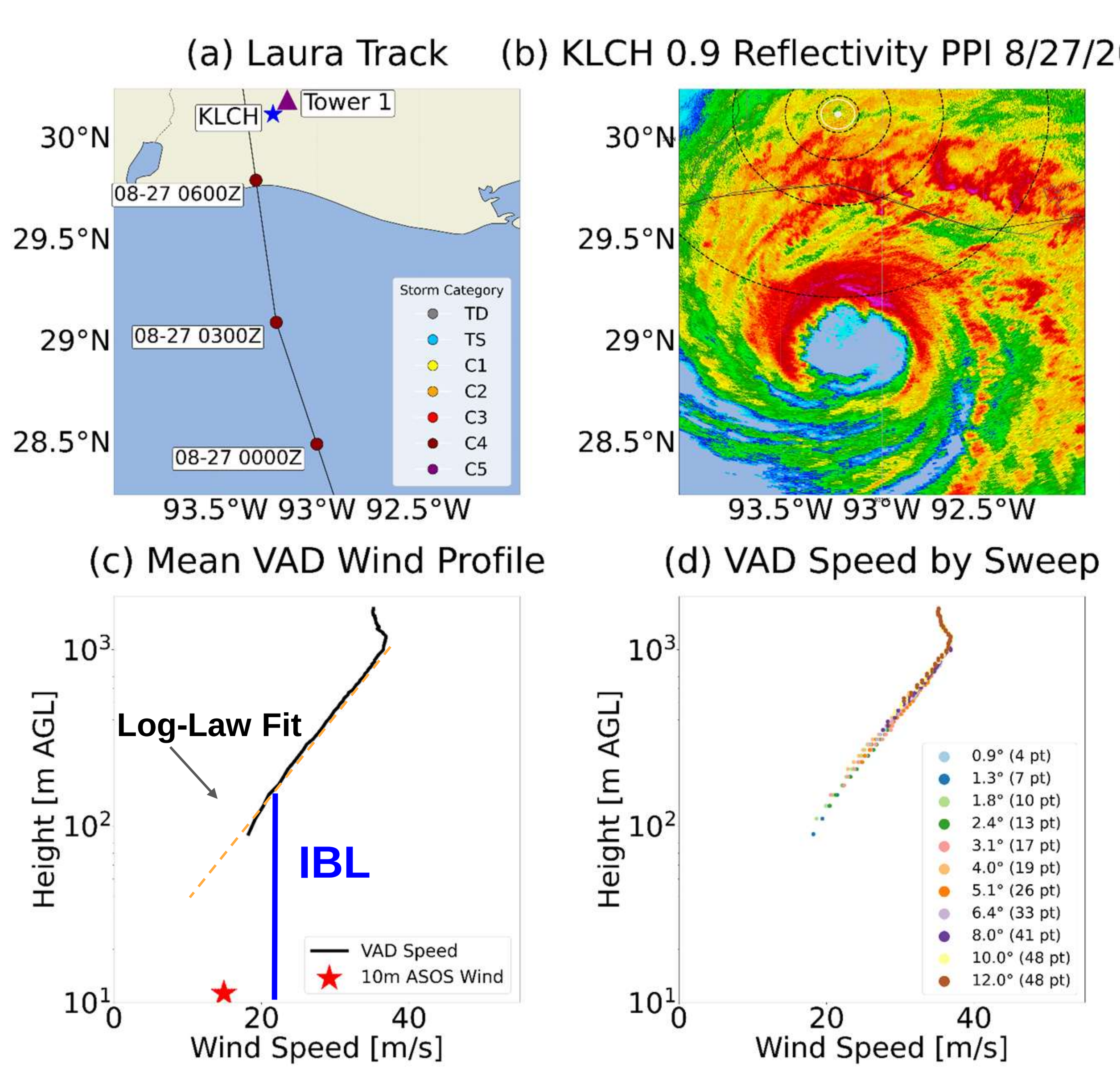

Fig 1. (a) Best Track of Laura, (b) 0.9° Radar Reflectivity 0239Z, (c-d) Mean VAD Profile and. VAD Speed by Sweep 0239Z.

Results

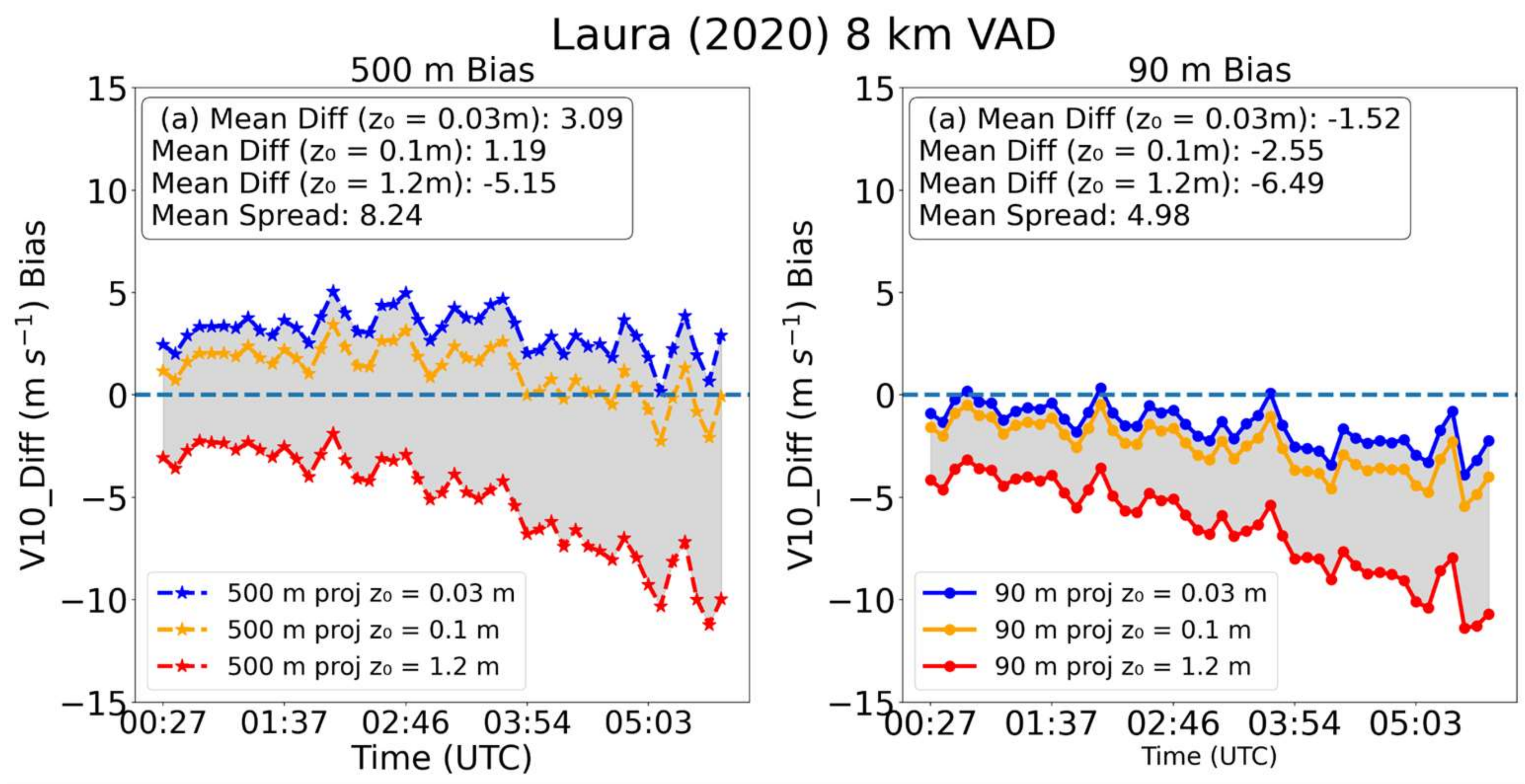

Fig 2. KLCH 10 m Projected vs ASOS Wind Speed Difference

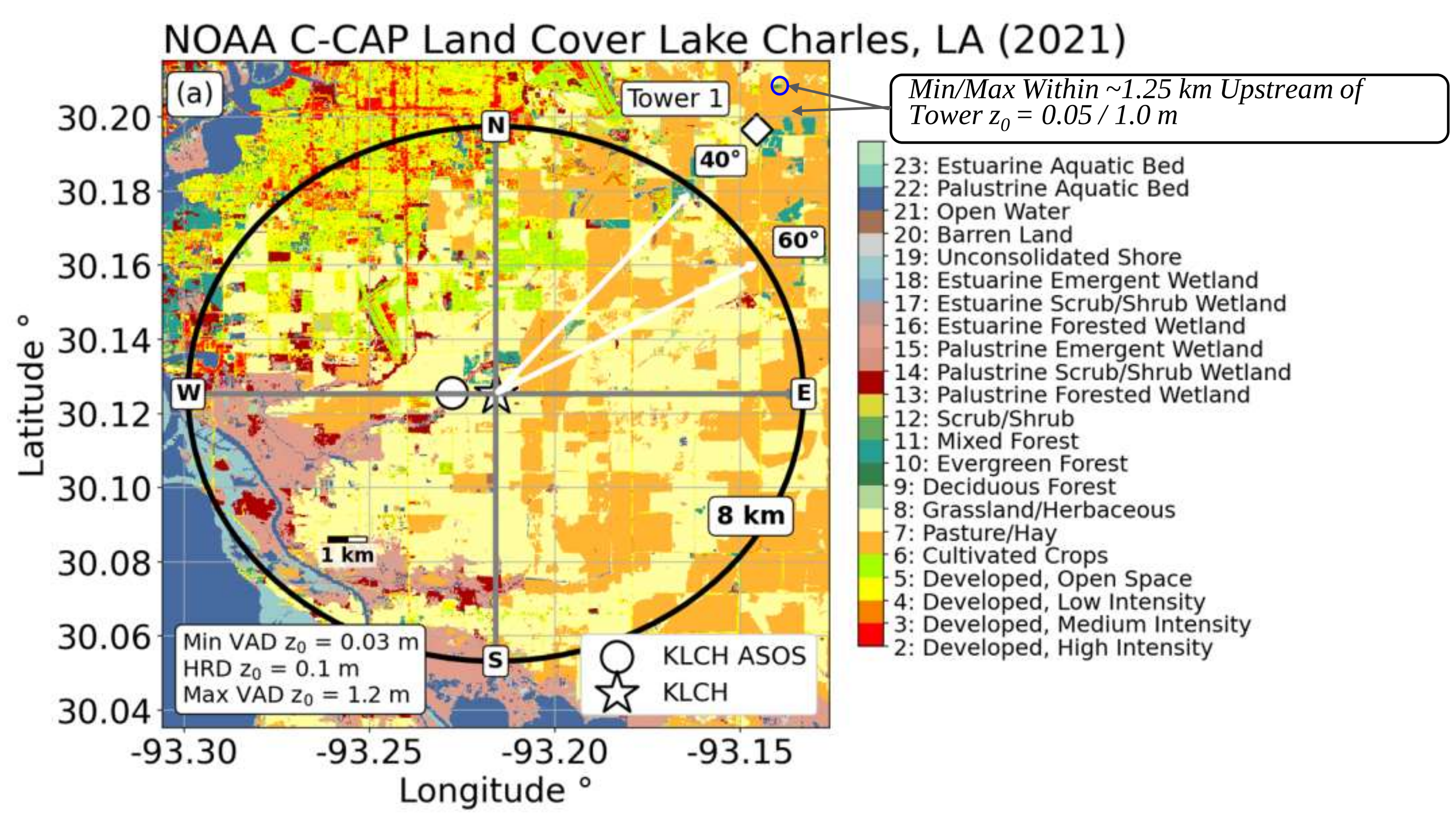

Fig 3. NOAA Land Use Classification 2021

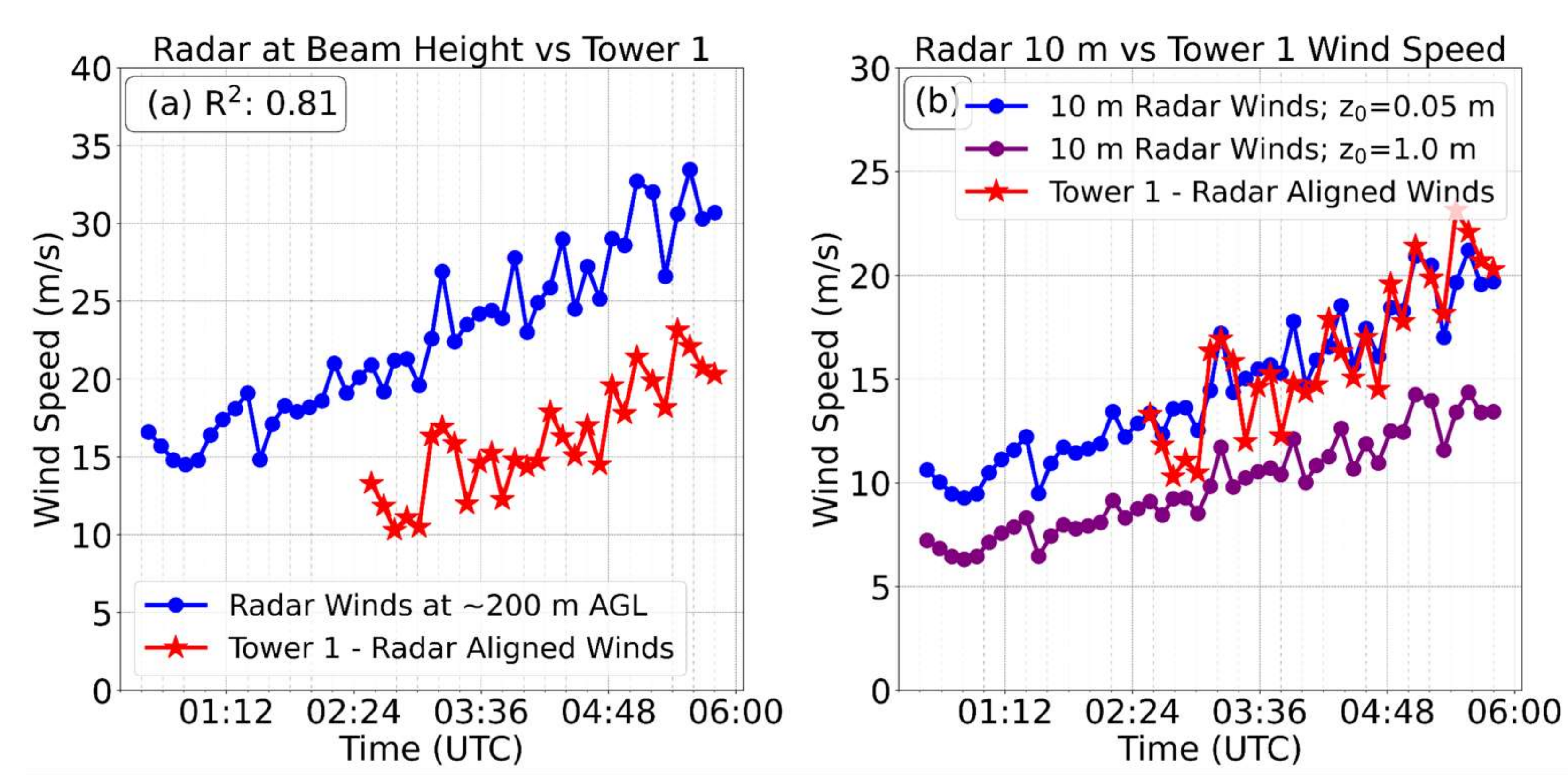

Fig 4. (a). KLCH Radar Gate Winds vs Tower 1 Winds in Radar Direction, (b) KLCH 10 m Projected vs Tower 1 Winds in Radar Direction

Conclusions

- Compared to the projection of 500-m winds, projecting 90-m VAD winds to 10-m AGL has less uncertainty to the spatial heterogeneity in upstream surface roughness.
- 500-m projection produced reasonable 10-m wind estimate using z_0=0.1 m, as suggested by the HRD website.
- FCMP Tower wind component in the radar direction correlates well with 200-m radar gate winds, indicating the existence of localized coherent turbulence structures.

References

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